SEAT NO:-31470020

DR. HOMI BHABHA STATE UNIVERSITY

THE INSTITUTE OF SCIENCE

DEPARTMENT OF BUSINESS AI

ACADEMIC YEAR-2026-27

STATISTICAL COMPUTING JOURNAL

NAME:- SMIT PATIL

CLASS:- SY B.Sc (BUSINESS AI)

SEAT NO:- 31470020

SUBJECT:- STATISTICAL COMPUTING JOURNAL

TEACHER SIGN:-_____

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DEPARTMENT OF BUSINESS AI CERTIFICATE

This is to certify that **Mr. SMIT PATIL** Exam Seat No./Roll **No. 31470020** B.Sc. Business AI (Semester III) has completed the practical work in the subject during the academic year 2026-27 under the guidance of **PROF. AKSHAY JOSHI** being the partial requirement for the fulfilment of the curriculum of Degree of Bachelor of Science in Business AI at The Institute of Science, Dr. Homi Bhabha State University.

Professor In-Charge **Coordinator**

Date: - ……. **College seal**

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4	To generate the Binomial PMF for n = 5 trials and probability of success p = 0.3, and visualize it using a bar chart.		
5	To generate the Poisson PMF for an average rate of $\lambda = 4$ complaints per day and plot the probability distribution.		
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10	To compute the regression intercept, slope and R-squared for Sales predicted by TV Spend, using ordinary least squares.		

Practical 1

Aim	To create and display a student-score data frame using Python and pandas.

Input	`import pandas as pd`
	
	`data = {`
	`"Student": ["Aarav", "Priya", "Rohit", "Sneha", "Kunal", "Divya"],`
	`"Score": [78, 85, 92, 67, 74, 88]`
	`}`
	
	`df = pd.DataFrame(data)`
	`print(df)`

Output	`Student Score`
	`0 Aarav 78`
	`1 Priya 85`
	`2 Rohit 92`
	`3 Sneha 67`
	`4 Kunal 74`
	`5 Divya 88`

Practical 2

Aim	To calculate the measures of central tendency mean, median and mode for the given sales data

Input	`import pandas as pd`
	
	`sales = pd.Series([42, 38, 51, 29, 46, 34, 61, 40, 37, 44])`
	
	`print("Mean:", sales.mean())`
	`print("Median:", sales.median())`
	`print("Mode:", sales.mode().tolist())`

Output	`Mean: 42.2`
	`Median: 41.0`
	`Mode: [29, 34, 37, 38, 40, 42, 44, 46, 51, 61]`

Practical 3

Aim	To calculate the sample variance and sample standard deviation for the given sales data.

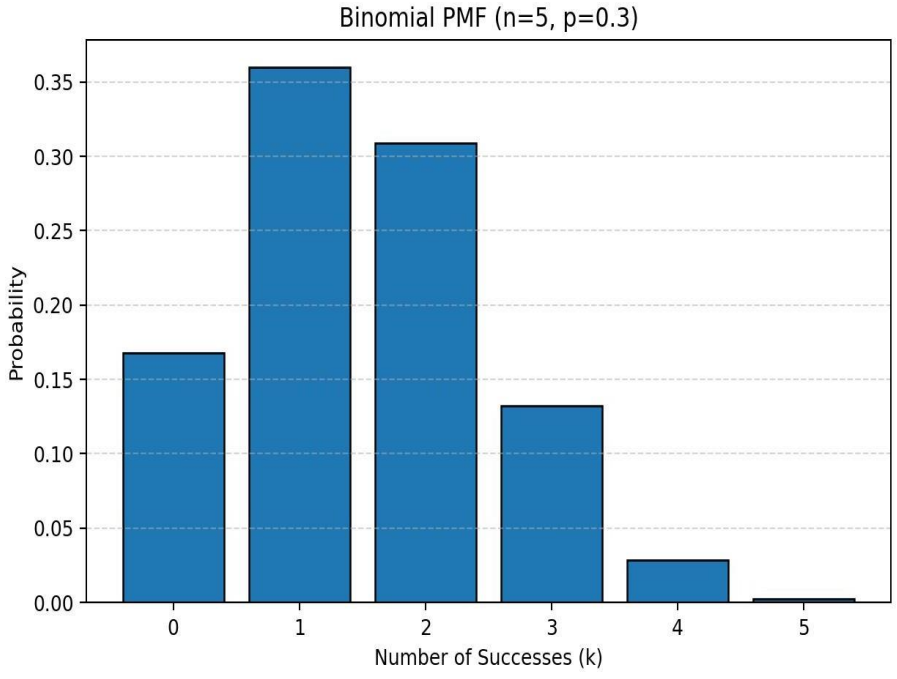
Input	`import pandas as pd`
	
	`sales = pd.Series([42, 38, 51, 29, 46, 34, 61, 40, 37, 44])`
	`variance = sales.var()`
	`std_dev = sales.std()`
	
	`print(f"Sample Variance: {variance:.2f}")`
	`print(f"Sample Std Dev: {std_dev:.2f}")`

Output	`Sample Variance: 82.18` `Sample Std Dev: 9.07`

Practical 4

Aim	To generate the Binomial PMF for n = 5 trials and probability of success p = 0.3, and visualize it using a bar chart.

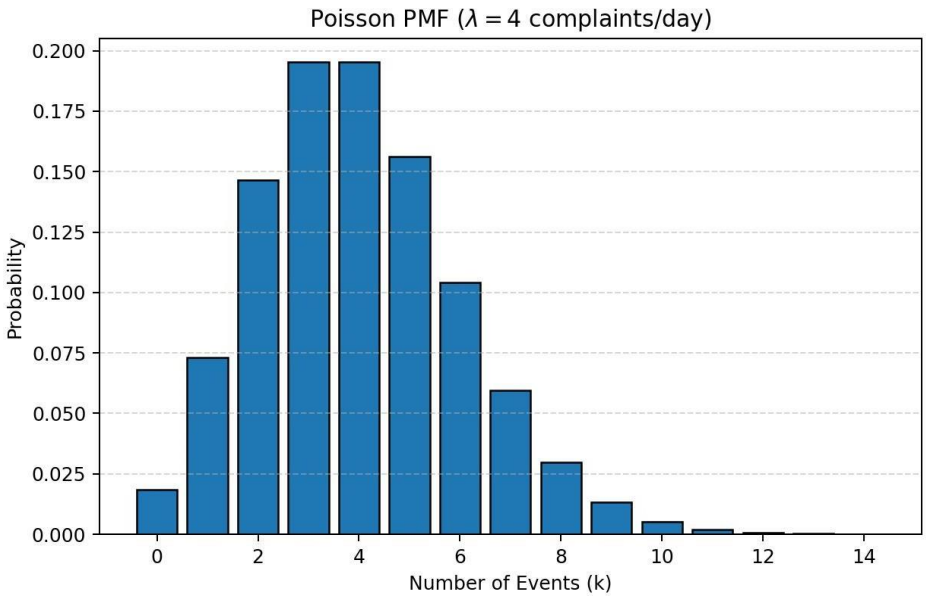
Input	`import numpy as np`
	`import scipy.stats as stats`
	`import matplotlib.pyplot as plt`
	
	`n, p = 5, 0.3`
	`k = np.arange(0, n + 1)`
	`pmf = stats.binom.pmf(k, n, p)`
	
	`for val, prob in zip(k, pmf):`
	`  print(f"P(X={val}) = {prob:.5f}")`
	
	`plt.bar(k, pmf, color="steelblue", edgecolor="black")`
	`plt.title("Binomial PMF (n=5, p=0.3)")`
	`plt.xlabel("Number of Successes (k)")`
	`plt.ylabel("Probability")`
	`plt.grid(axis='y', linestyle='--', alpha=0.7)`

Output	`P(X=0) = 0.16807`
	`P(X=1) = 0.36015`
	`P(X=2) = 0.30870`
	`P(X=3) = 0.13230`
	`P(X=4) = 0.02835`
	`P(X=5) = 0.00243`
	 

Practical 5

Aim	To generate the Poisson PMF for an average rate of λ = 4 complaints per day and plot the probability distribution.

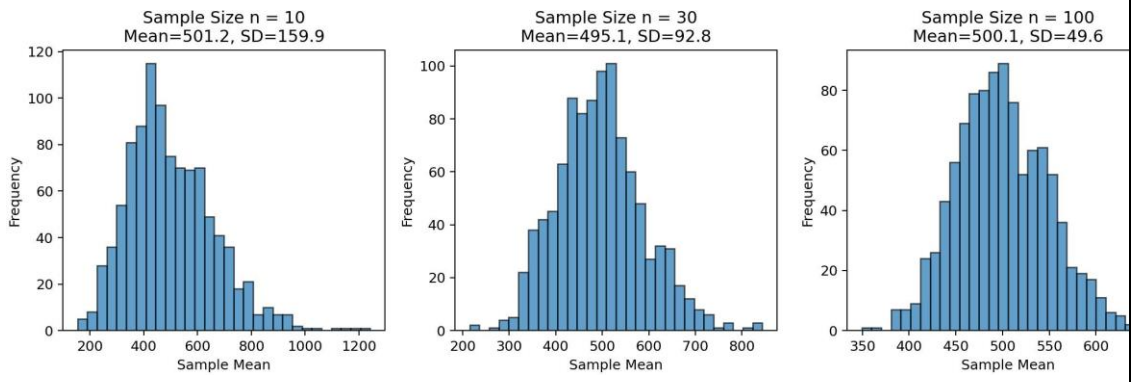
Input	`import numpy as np`
	`import scipy.stats as stats`
	`import matplotlib.pyplot as plt`
	
	`lam = 4`
	`k = np.arange(0, 15)`
	`pmf_poisson = stats.poisson.pmf(k, lam)`
	
	`print(f"P(X=3) = {stats.poisson.pmf(3, lam):.5f}")`
	
	`plt.bar(k, pmf_poisson, color="orange", edgecolor="black")`
	`plt.title(r"Poisson PMF ($\lambda=4$ complaints/day)")`
	`plt.xlabel("Number of Events (k)")`
	`plt.ylabel("Probability")`
	`plt.grid(axis='y', linestyle='--', alpha=0.7)`
	`plt.show()`

Output	`P(X=3) = 0.19537` 

Practical 6

Aim	To demonstrate how the distribution of sample means becomes more concentrated and approximately normal as sample size increases, using an exponential (non-normal) population.

Input	`import numpy as np`
	`import matplotlib.pyplot as plt`
	
	`np.random.seed(42)`
	`# Generating non-normal skewed population (Exponential)`
	`population = np.random.exponential(scale=500, size=100000)`
	
	`sample sizes = [10, 30, 100]`
	`fig, axes = plt.subplots(1, 3, figsize=(15, 4))`
	
	`for i, n in enumerate(sample_sizes):`
	`sample_means = [np.random.choice(population, size=n, replace=False).mean() for _ in range(1000)]`
	`axes[i].hist(sample_means, bins=30, color="teal", edgecolor="black", alpha=0.7)`
	`axes[i].set_title(f"Sample Size n = {n}\nMean={np.mean(sample_means):.1f}, SD={np.std(sample_means):.1f}")`
	`axes[i].set_xlabel("Sample Mean")`
	
	`plt.tight_layout()`
	`plt.show()`

Output	`Sample Size n = 10: Mean=501.2, SD=159.9`
	`Sample Size n = 30: Mean=495.1, SD=92.8`
	`Sample Size n = 100: Mean=500.1, SD=49.6`
	
	 

Practical 7

Aim	To compute 90%, 95% and 99% confidence intervals for the mean weight using the t-distribution

Input	`import pandas as pd`
	
	`sales = pd.Series([42, 38, 51, 29, 46, 34, 61, 40, 37, 44])`
	
	`print("Mean:", sales.mean())`
	`print("Median:", sales.median())`
	`print("Mode:", sales.mode().tolist())`

Input	90% CI: (248.50, 251.50) 95% CI: (248.15, 251.85) 99% CI: (247.35, 252.65)

Practical 8

Aim	To test whether the mean bottle volume differs significantly from the hypothesized population mean of 500 ml.

Input	`import numpy as np`
	`from scipy import stats`
	
	`bottles = np.array([498, 502, 497, 501, 499, 503, 496, 500])`
	`t_stat, p_val = stats.ttest_1samp(bottles, popmean=500)`
	
	`print(f"t-statistic: {t_stat:.3f}")`
	`print(f"p-value: {p_val:.4f}")`

Output	`t-statistic: -0.577` `p-value: 0.5818`

Practical 9

Aim	To determine whether the mean performance of Shift A differs significantly from Shift B.

Input	`import numpy as np`
	`from scipy import stats`
	
	`shift a = np.array([45, 48, 42, 50])`
	`shift b = np.array([52, 55, 50, 58])`
	
	`t_stat, p_val = stats.ttest_ind(shift_a, shift_b)`
	`print(f"t-statistic: {t_stat:.3f}")`
	`print(f"p-value: {p_val:.4f}")`

Output	

```
t-statistic: -2.490
p-value: 0.0471
```

Practical 10

Aim	To compute the regression intercept, slope and R-squared for Sales predicted by TV Spend, using ordinary least squares.

Input	`import numpy as np`
	`ort pandas as pd`
	`ort statsmodels.api as sm`
	
	`= pd.DataFrame({`
	`"TV Spend": [100, 150, 200, 250, 300],`
	`"Sales": [150, 200, 230, 280, 320]`
	
	`dd constant intercept column for OLS`
	`sm.add_constant(df["TV_Spend"])`
	`df["Sales"]`
	
	`el = sm.OLS(y, X).fit()`
	
	`nt(f"Intercept (beta_0): {model.params['const']:.2f}")`
	`nt(f"Slope (beta_1):  {model.params['TV_Spend']:.2f}")`
	`nt(f"R-squared:  {model.rsquared:.4f}")`

Output	`Intercept (beta 0): 68.00`
	`Slope (beta_1): 0.84`
	`R-squared: 0.9955`

Output	